		Hence acceleration on trailer = 400/400 = 1 m s Net force on car = ma = 1200 x 1 = 1200 N
5	C	Relative speed of approach: $u_1 - u_2 = v_2 - v_1 = 6 - (-1) = 7$. Only B and C has $v_2 - v_1 = 7$ Conservation of linear momentum: Initial total momentum = $6M - m$ where M is mass of A and m mass of B For option B: $6M - m = -1M + 6m \rightarrow 7M = 7m$ not possible as $M \neq m$ For option C: $6M - m = 1.3M + 8.3m \rightarrow 4.7M = 9.3m$ ok as $M > m$
6	C	Upthrust on ice cube = weight of water displaced